# Sahaj Bhadja

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## Summary

Software engineer with **2+ years of experience** building research and production software across AI/ML, data pipelines, and Linux-backed platforms, turning prototypes into reliable systems with **CI/CD, multi-environment deployment, and collaborative Git workflows**.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

## Experience

**USC Dornsife** **LA, California**
*Software Engineer* *September 2025 – Present*

- Architected a **multi-tenant AI education platform** on **Azure SQL, PostgreSQL, and Blob Storage** for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through schema normalization, indexing, and query optimization.
- Built **FastAPI**-based async orchestration and ETL-style integration workflows across relational stores and cloud storage, processing **1K+ AI jobs** and **5K+ monthly integration jobs** with **99%+ reliability**.

**Grad8** **LA, California**
*Software Engineer Intern* *June 2025 – August 2025*

- Developed a **Flask + React** grading platform using multimodal AI and OCR to evaluate **10K+ text and handwritten submissions**, supporting secure PDF/DOCX/ZIP ingestion and instructor-facing review workflows.
- Designed concurrent backend pipelines with **ThreadPoolExecutor**, **MySQL**, and **GitHub Actions CI/CD**, handling **10K+ grading requests/day** for **150+ instructors** while cutting feedback turnaround time by **60%**.

**USC AD Labs** **Los Angeles**
*ML Research Assistant* *January 2025 – May 2025*

- Built **PyTorch** simulation and model-training pipelines over **100K+ trajectory logs** and **10K autonomous-driving episodes**, preparing HD-map and agent-context features for offline reinforcement learning in Linux-based research workflows.
- Designed a spatial-temporal encoder for Decision Transformer planning and closed-loop evaluation, improving long-horizon planning by **18%** and planning stability by **15%** while collaborating with faculty and research teammates.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*

- Built a cross-platform **microservice** data-sharing platform and automated nightly build, install, and validation flows across **Windows, Linux, and macOS** VMs using **Jenkins, PyTest, and Mockito**, improving response times by **30%**.
- Reworked **AWS Glue**, **S3**, SQL, and API workflows for production ETL and documented developer-facing test procedures, increasing pipeline efficiency by **68%**, reducing technical debt by **20%**, and lowering storage costs by **15%**.

## Projects

**CPU-Scheduling Simulator** Live | *C++, React.js, EJ-2*

- Implemented **FCFS, SJF, SRT, RR, priority scheduling, and MLQ** with a React-based UI and Gantt visualization, validating behavior across **1K+ simulated process combinations**, reducing debugging time by **20%** and average turnaround time by **25%**.

**CaptionCraft - Image Caption Generator** Live | *Transformers, PyTorch, Flask*

- Benchmarked **BLIP, GIT, and ViT-GPT-2** on a **10K image / 100K caption dataset**, improving caption accuracy by **15%** (**81% to 96%**) while building Flask inference APIs for scalable real-time caption generation.

**Chat2SQL** Repo | *Java, TypeScript, SQL, Docker*

- Built an AI-powered SQL workspace on top of the Chat2DB project, supporting **10+ databases** and cross-platform development on **Windows, macOS, and Linux**, accelerating schema exploration and structured query generation with **82%** validation accuracy.

## Skills

**Languages**: Python, C/C++, Java, SQL, JavaScript/TypeScript, Shell
**Research & Backend**: PyTorch, Transformers, Flask, FastAPI, REST APIs, OOP, Async Processing
**Data & Infrastructure**: PostgreSQL, MySQL, Azure SQL, AWS (S3, RDS, Glue), Docker, Kubernetes, Linux
**Engineering**: Git/GitHub, CI/CD, Jenkins, GitHub Actions, Testing Automation, Distributed Systems, Data Pipelines